

SPOTIFY MUSIC RECOMMENDERE WEB APP

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1. Introduction

With so much music available on streaming platforms, finding new songs that match our taste can be overwhelming. That's why I developed this Spotify Music Recommendation System, that makes discovering music easier and more personalized. Using machine learning and AI techniques, my system suggests relevant songs based on user input. It combines clustering, and KNN algorithms to generate recommendations that are more tailored to individual preferences all neatly displayed on a custom and well structure web app.

2. Background

Based on research that I've conducted, I have found that recommender systems are widely used in e-commerce, entertainment, and media industries to enhance user experiences by suggesting relevant content. Platforms like Spotify, Apple Music, and YouTube Music use machine learning to curate personalized playlists. A recent project released by Spotify, their "Spotify DJ" is a new and innovative concept to the platform and amongst its competitors. So, I attempted to create a project that aims to replicate and enhance such a system using a Spotify dataset.

3. Problem statement

With millions of songs available on the streaming platform, users can find it difficult to find new music that aligns with their preferences. Existing recommendation systems often rely on popularity metrics rather than personalized content-based analysis, leading to suboptimal music suggestions. This project seeks to improve recommendation accuracy by integrating fuzzy search, multi-keyword search, and cluster-based recommendations which should help closely align more with similar songs that users are interested in listening to.

4. Project objectives

- Implement an efficient search tool for better song retrieval and provide accurate results.
- Provide personalized recommendations using collaborative and content-based filtering.
- Introduce a clickable album feature, allowing users to explore all tracks from an album.
- Implement popularity-based fallback recommendations when no search match is found.
- Develop visualization tools to analyse trends in recommended music.

5. Dataset / Data warehouse details

The dataset was found on Kaggle -

<https://www.kaggle.com/datasets/gauthamvijayarai/spotify-tracks-dataset-updated-every-week>,

This dataset contains over 62000 songs and has regular updates. It includes track metadata (track name, artist name, album, release year), Audio features (danceability, energy,

valence, tempo, etc.), popularity scores (measuring how well a track is performing) and language labels (useful for filtering recommendations by language)

6. Interesting insights

What I have noticed from the dataset was the popularity trends that differ from song to song. The popularity trends can even vary from genre to language. I also noticed how the release year does have a significant impact on the popularity scores of certain tracks.

7. Data pre-processing

When preparing my dataset I took the following steps :

- Removed missing values in critical columns (track name, artist name, etc.).
- Normalized numerical features using StandardScaler to improve model performance.
- Converted popularity scores to integers for efficient processing.
- Created a user-item matrix for collaborative filtering.

8. Measuring performance

- Root Mean Square Error (RMSE) for collaborative filtering.
- Cosine Similarity Scores to assess similarity-based recommendations.
- User Testing & Feedback to gauge recommendation relevance.

9. Algorithms applied

An overview of selected approach(es) / machine learning algorithm(s) that have been applied to the data. More than one algorithm or approach may be applied for the same purpose, i.e., for the sake of comparison to select the best performing algorithm or they may be applied for different reasons, i.e., to gather different insights.

For each machine learning algorithm provide an overview of the algorithm, and why it was selected. For all algorithms, provide code examples of how the models were created, trained, and tested/validated. Comparative results, such as outputs from different contrasting metrics, should be presented in tables or visualised.

Sub-headings help structure your report document and make it easier for the reader to follow. It is advised that you create sub-headings for the algorithms (models) which you create, as well as a separate one to compare and/or contrast their performance. This is if you select more than one approach or algorithm. If only one approach or algorithm is selected, you need only highlight the related approach or algorithm.

9.1 K-Means

I used it to group songs based on audio features. It also helped in providing recommendations from similar musical styles.

9.2 K-Nearest Neighbors (KNN).

I used KNN to find similar tracks based on user-item interactions. This ensured personalized recommendations based on user preferences.

9.3 Hybrid Recommendation Approach

Combined collaborative filtering (KNN) and content-based filtering (clustering) for improved accuracy.

10. Discussion

Building this recommendation system taught me a lot about how different filtering methods can improve music discovery. By adding fuzzy search and multi-keyword search, I made it easier for users to find tracks, even with typos or incomplete queries. The clickable album feature was another useful addition, allowing users to explore an entire album with one click.

One challenge I encountered was the cold-start problem, where new or unpopular songs lacked enough data for accurate recommendations. Additionally, popular tracks tended to dominate recommendations, which sometimes limited diversity. To address these, I used popularity-based fallback recommendations to ensure users always received relevant suggestions.

Overall, the hybrid approach combining KNN (collaborative filtering) and K-Means clustering (content-based filtering) worked well to generate personalized and diverse recommendations. In the future, I would explore integrating deep learning models or real-time user feedback to further refine the system.

11.Conclusion

This project successfully developed a Spotify Music Recommendation System with enhanced search flexibility, personalized suggestions, and interactive features. The objectives were met through KNN and clustering, ensuring high-quality recommendations. The use of visual analytics provided deeper insights into user preferences. Future work could focus on real-time recommendations and user behavior tracking to refine personalization.

References

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